

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi-110067

Mid Semester Exam March 2024

Program:- MCA (II Semester)

Course:- Computer Networks (CS408)

Max Marks:- 20

Time:- 90 Minutes

Note: Attempt all the questions, assume suitable data if missing.

- Q. 1.** Discuss different characteristics of Data Link layer and Transport layer of the OSI Model. [4]
- Q. 2.** What is the need of Logical Address? Discuss different types of address used in networks. [4]
- Q. 3.** A sine wave is offset $1/9$ cycle with respect to time 0. What is its phase in degrees and radians? [1]
- Q. 4.** The power of a signal is 10 milliwatt and power of the noise is 0.5 microwatt, what is the value of the SNR_{dB} ? [2]
- Q. 5.** We need to send 272kbps over a noiseless channel with a bandwidth of 20kHz, calculate the how many signal levels do we need. Can we send data with above mentioned bit rate or not? Justify your answer. [3]
- Q. 6.** The attenuation of a signal is -13dB. What was the original signal power if its current power 3Watt? [2]
- Q. 7.** Which signal has a wider bandwidth, a sine wave with frequency of 150Hz or a sine wave with a frequency of 250Hz? [1]
- Q. 8.** Suppose a signal with 250milliwatts power passes through from point X1 to point X5. From point X1 to point X5 at the even points (X2, X4) the signal is attenuated with 4db and at odd point (X3) the signal is amplified with -6dB. What will be the power of the signal at point X5 in milliwatts? [3]

End

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi-110067

End Semester Exam May 2024

Program:- MCA (II Semester)

Course:- Computer Networks (CS408)

Time:- 180 Minutes

Student Roll No.....

Max Marks:- 50

Exam date: 20-05-2024

Note: Attempt all the questions, assume suitable data if missing.

Q.1.

- A. An image is 1024×768 pixel with 6bytes/pixel. Assume the image is uncompressed. How long does it take to transmit it over a 56-kbps modem channel? Over a 1-Mbps cable modem? Over 10Mbps Ethernet? Over 100Mbps Ethernet? [2 marks]
- B. We need a dataword of at least 16bits. Find the values of k and n in the Hamming code $C(n, k)$ with $d_{min}=3$. [1 mark]
- C. Discuss the transmission Impairment in details. [3 marks]
- D. What is the need of Multiplexing? Explain the FDM multiplexing techniques with suitable example. [4 marks]

Q.2.

- A. Assume that, in a Stop-and Wait ARQ system, the bandwidth of the line is 10Mbps and 1 bit takes 15ms to take a round trip. If the system data frames are 79000 bits in length, what is the utilization percentage of the link? [2 marks]
- B. A pure ALOHA network transmits 450bit frames on a shared channel of 250kbps. What is the requirement to make this frame collision-free? [2 marks]
- C. Discuss the Go-Back-N Automatic Repeat Request Protocol with suitable example. [3 marks]
- D. A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system produces [3 marks]
 - 1000 frames per second
 - 500 frames per second
 - 250 frames per second.

Q.3.

- A. An ISP is granted a block of address starting with 172.100.0.0/16. The ISP needs to distribute these addresses to three groups of customers as follows: [6 marks]
 - a) The first group has 128 customers; each needs 64 address.
 - b) The second group has 128 customers; each needs 128 addresses.
 - c) The third group has 64 customers; each needs 256 addresses.

Design the sub-blocks and find out how many addresses are still available after these allocations.

- B.** What are the advantage of the IPv6 as compared to the IPv4 protocols? Discuss the Packet format of the IPv6 protocol. [4 marks]

Q.4.

- A. How connection is established in the TCP protocol? Discusses the TCP connection procedure with suitable example. [3 marks]
- B. Define Congestion in the Network. Discuss any two congestion control techniques in detail. [4 marks]
- C. Consider a network with five nodes A, B, C, D, and E. There are undirected edges (nodes are connected via communication edges). The cost of edges is given as follows: From node A to node B is 3, Node A to node C is 4, Node A to D is 5, Node B to node C is 2, Node B to node E is 4, and Node C to Node E is 2. Use the step wise step procedure of the Distance vector routing algorithm and find the least cost between every two nodes. [3 marks]

Q.5. Write short note on following: [10 marks]

- Electronic Mail
- Uniform Resource Locator
- HTTP protocol
- DNS

End

School of Computer and Systems Sciences
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New Delhi-110067

Mid Semester Exam March 2024

Program:- MCA (II Semester)

Course:- Computer Networks (CS408)

Time:- 90 Minutes

Student Roll No....14.....

Max Marks:- 20

Exam date: 13-03-2024

Note: Attempt all the questions, assume suitable data if missing.

- Q.1.** What is the need of Computer Networks? Discuss different types of transmission technologies used in networks. [4]
- Q.2.** Define the role of each layer of TCP/IP model with an example of supporting protocol. [4]
- Q.3.** It is desired to send a sequence of computer screen image over an Optical Fiber. The screen size is 525×700 pixels, each pixel being 12 bits. There are 60-screen image per second. How many bandwidth is needed in bps and what is wavelengths needed in microns. [3]
- Q.4.** What will be the bandwidth of a low pass channel? If it's available bit rate of a channel is 750Mbps. [1]
- Q.5.** If a network with bandwidth is, 60kbps and we want to pass data with an average of 6000 lines per minutes (each line contains 80 characters and each character is represented by 8bits). Identify weather we can send number of bits or not, justify your answer. [2]
- Q.6.** In a digital transmission, the receiver clock is 0.4 percent slower than the sender clock. How many fewer (less) bits per second does the receiver receive if the data rate is 2mbps? [1]
- Q.7.** Suppose a signal with 300milliwatts power passes through from point P1 to point P5. From point P1 to point P5 at the even points (P2, P4) the signal is amplified with 5db and at odd point (P3) the signal is attenuated with -6dB. What will be the power of the signal at point P5 in microwatts? [3]
- Q.8.** If the signal levels are 8 and the maximum bit rate is 24mbps what will be Signal to Noise Ratio and bandwidth of the channel? [2]

End

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School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi-110067

End Semester Exam May 2024

Program:- MCA (II Semester)

Course:- Computer Networks (CS408)

Max Marks:- 50

Time:- 180 Minutes

Exam date: 20-05-2024

Student Roll No.....24.....

Note: Attempt all the questions, assume suitable data if missing.

Q.1.

- A. An image is 1024×768 pixel with 6bytes/pixel. Assume the image is uncompressed. How long does it take to transmit it over a 56-kbps modem channel? Over a 1-Mbps cable modem? Over 10Mbps Ethernet? Over 100Mbps Ethernet? [2 marks]
- B. We need a dataword of at least 16bits. Find the values of k and n in the Hamming code $C(n, k)$ with $d_{min}=3$. [1 mark]
- C. Discuss the transmission Impairment in details. [3 marks]
- D. What is the need of Multiplexing? Explain the FDM multiplexing techniques with suitable example. [4 marks]

Q.2.

- A. Assume that, in a Stop-and Wait ARQ system, the bandwidth of the line is 10Mbps and 1 bit takes 15ms to take a round trip. If the system data frames are 79000 bits in length, what is the utilization percentage of the link? [2 marks]
- B. A pure ALOHA network transmits 450bit frames on a shared channel of 250kbps. What is the requirement to make this frame collision-free? [2 marks]
- C. Discuss the Go-Back-N Automatic Repeat Request Protocol with suitable example. [3 marks]
- D. A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system produces [3 marks]
- 1000 frames per second
 - 500 frames per second
 - 250 frames per second.

Q.3.

- A. An ISP is granted a block of address starting with 172.100.0.0/16. The ISP needs to distribute these addresses to three groups of customers as follows: [6 marks]
- a) The first group has 128 customers; each needs 64 address.
 - b) The second group has 128 customers; each needs 128 addresses.
 - c) The third group has 64 customers; each needs 256 addresses.

Design the sub-blocks and find out how many addresses are still available after these allocations.

B. What are the advantage of the IPv6 as compared to the IPv4 protocols? Discuss the Packet format of the IPv6 protocol. [4 marks]

Q.4.

A. How connection is established in the TCP protocol? Discusses the TCP connection procedure with suitable example. [3 marks]

B. Define Congestion in the Network. Discuss any two congestion control techniques in detail. [4 marks]

C. Consider the following information of the nodes and cost from one node to another node is given in the below figure: [3]

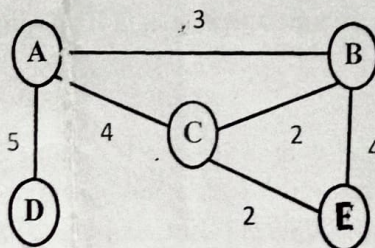


Fig. Connected nodes with their cost of edges

Use the step wise step procedure of the Distance vector routing algorithm and find the least cost between every two nodes.

Q.5. Write short note on following:

[10 marks]

- Electronic Mail
- Uniform Resource Locator
- HTTP protocol
- DNS

End*

School of Computer & Systems Sciences
CS - 112, Computer Networks
MCA-Mid Semester Examination-2014

Time: 2 hours

Max. Marks: 25

Note: Attempts all questions. Each question carries equal marks.

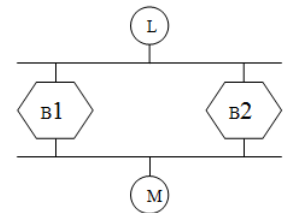
Q1. Suppose we want to transmit the 8-bit message $M = 11100011$ and protect it from errors using the generator polynomial $P(x) = x^5 + x^4 + x + 1$.

- (i) Encode the message M using the generator polynomial and give the codeword (T : Frame to be transmitted).
- (ii) Suppose the channel introduces an error pattern $E(x) = x^9 + x^6 + x^3$ in T during the transmission. What is received at receiver? Can the error be detectable at the receiver?

Q2. (a) What is p -Persistent CSMA protocol? How p -Persistent protocol behaves (in terms of collision and delay) for small value of p under heavy load and under low load?

(b) What are the corresponding layers of the following entities: gateway, hub, repeater, router, and switch? Explain the difference between hub and switch.

Q3. For the network given in figure, two transparent bridges $B1$ and $B2$ are connected form a loop and do not implement the spanning tree algorithm. What will happen if host M send to host L ?



Q4. Consider all stations, numbered 1 through 2^n ($n \geq 2$), are contending for the use of a shared channel by using the adaptive tree walk protocol. If all the station whose addresses are $4, 8, 16, 32, 64, \dots$ suddenly becomes ready at once, how many bit slots (in terms of n) are needed to resolve the contention?

Q5. Consider the subnet of Fig 2. Distance vector routing is used, and the routing vectors are established for routers A, B, C, D, E , and F (as given in Table 1): and then the $C-E$ link fails. Give

- (i) The routing tables of A, B, D , and F after C and E have reported the news to their neighbors.
- (ii) The routing tables of A and D after their next mutual exchange.

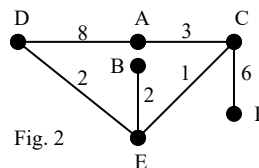


Table-1						
Routing table at router	Cost to reach router					
	A	B	C	D	E	F
A	0	6	3	6	4	9
B	6	0	3	4	2	9
C	3	3	0	3	1	6
D	6	4	3	0	2	9
E	4	2	1	2	0	7
F	9	9	6	9	7	0

School of Computer & Systems Sciences
Jawaharlal Nehru University
CS - 505, Computer Networks and Distributed Systems
MCA III – Semester
Mid Semester Examination

Time: 120 Minutes

Max. Marks: 20

Note: Attempts all questions.

1. What is Bit-Map protocol? Briefly analyze the performance (efficiency) of this protocol at low and high load. How long does a station have to wait in the worst case before it can start transmitting its frame over a LAN that uses this protocol? (5)

2. (a) Explain the transparent bridges. How they operate? (2 + 3)

- (b) Consider hosts S, T, U, V, W, X, Y, Z and transparent bridges $B1, B2, B3, B4$ with initially empty hash tables as in figure 1. Show how the hash tables of different bridges change after each of the following events happen in sequence, first (i) then (ii) and so on.

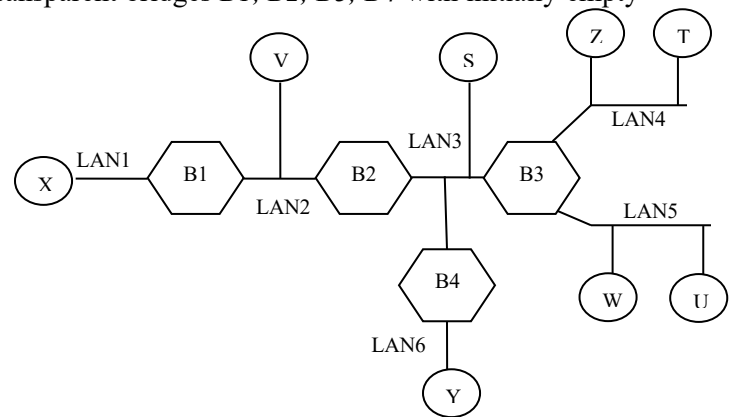


Figure 1

- (i) Suppose X sends to Z . Which bridges learn where X is? Does Y 's network interface see this frame?
(ii) U sends to X . Which bridges learn where U is? Does Y 's network interface see this frame?
(iii) T sends to V . Which bridges learn where T is? Does X 's network interface see this frame?
(iv) U moves to $LAN6$. Which bridges learn where U is? And then U sends to T . Which bridges learn where U is? Does W 's network interface see this frame?

3. (a) Consider the subnet of figure 2. Distance vector routing is used, and the following vectors have just come in to router C : from B : $(5, 0, 8, 12, 6, 2)$; from D : $(16, 12, 6, 0, 9, 10)$; and E : $(7, 6, 3, 9, 0, 4)$. The measured delays to B, D , and E , are $6, 3$, and 5 , respectively. What is C 's new routing table? Give both the routing line to use and the expected delay. (2.5 + 2.5)

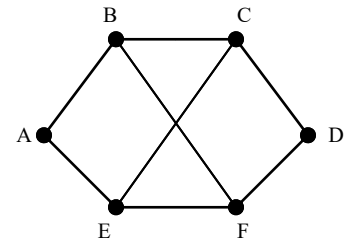


Figure 2

- (b) What are the five advantages of Packet Switching over Circuit Switching?

4. (a) A CRC is constructed to generate a 4-bit FCS (Frame Check Sequence) for an 11-bit message $D = 10010011011$. The generator polynomial is $P(x) = x^4 + x + 1$.

- (i) Encode the message D using the generator polynomial and give the codeword (T : Frame to be transmitted). (3 + 2)

- (ii) Suppose the channel introduces an error pattern $E(x) = x^{14} + x^{11} + x^{10}$ in T during the transit. What is received? Can the error be detected?

- (b) What are the three frame types supported by HDLC (High Level Data Link Control) protocol? Describe each.